Enterprise Data Management Plan for Apple Fitness+

Improving Customer Engagement and Retention

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4.3 Final Project: EDM Plan Overview

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Executive Summary and EDM Plan

Apple Fitness+ is a subscription fitness and wellness service offering guided workouts and meditations across Apple devices. Apple has publicly described personalized plans and a broad workout library (Apple, 2025a, 2025b). This proposal recommends a governed, customer-focused data warehouse to help management evaluate engagement and subscription outcomes. Apple does not publicly disclose its complete internal EDM architecture; potential data silos and inconsistent reporting are therefore discovery hypotheses, not claims about Apple's actual systems.

The plan applies customer relationship management concepts from Chapter 8 of Kimball and Ross (2013) and the broader DW/BI lifecycle in Chapter 17. The proposed stack is Airbyte Cloud for ingestion, Azure Data Lake Storage for a restricted raw zone, Microsoft Fabric Warehouse for integrated dimensional data, and Power BI for analytics. The pilot uses only approved, minimized, pseudonymous data and excludes sensitive health information that is not necessary for the business questions.

Dimensional Schema and Architecture

The schema defines facts, dimensions, keys, relationships, and grain; architecture describes platforms and data movement. FactWorkoutActivity has a grain of one eligible workout session per row, with date, customer, workout, duration, and completion measures. FactSubscriptionActivity has one subscription lifecycle event per row, such as a start, renewal, or cancellation. Dimensions include DimCustomer (pseudonymous key and approved segment), DimDate, DimWorkout, and DimSubscriptionPlan. Defining grain before selecting measures helps avoid ambiguous counts and double-counting (Kimball Group, 2003, 2015).

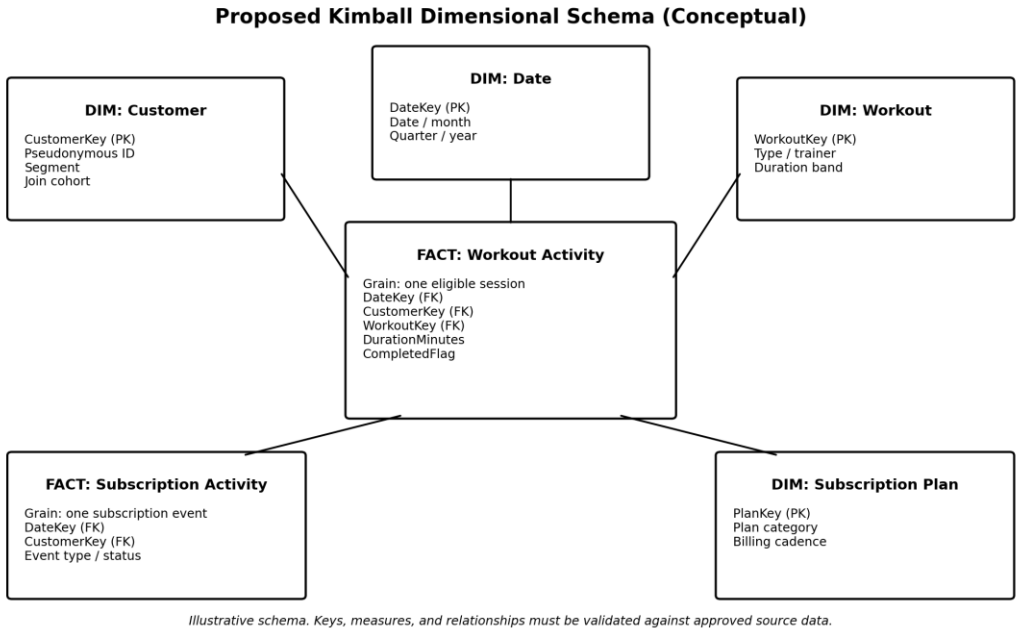


Figure 1

Proposed Kimball Dimensional Schema

Note. Conceptual model; validate keys, measures, and relationships against approved source data.

Approved extracts or APIs flow through Airbyte Cloud into a restricted Azure Data Lake Storage raw zone, preserving source timestamps and metadata. Quality checks validate schemas, required fields, duplicate keys, and referential integrity. Standardized transformations load conformed facts and dimensions into Microsoft Fabric Warehouse. Power BI serves governed engagement, retention, content-mix, and data-freshness metrics. Microsoft's licensing guidance should be reviewed to choose the right capacity and sharing model (Microsoft, n.d.).

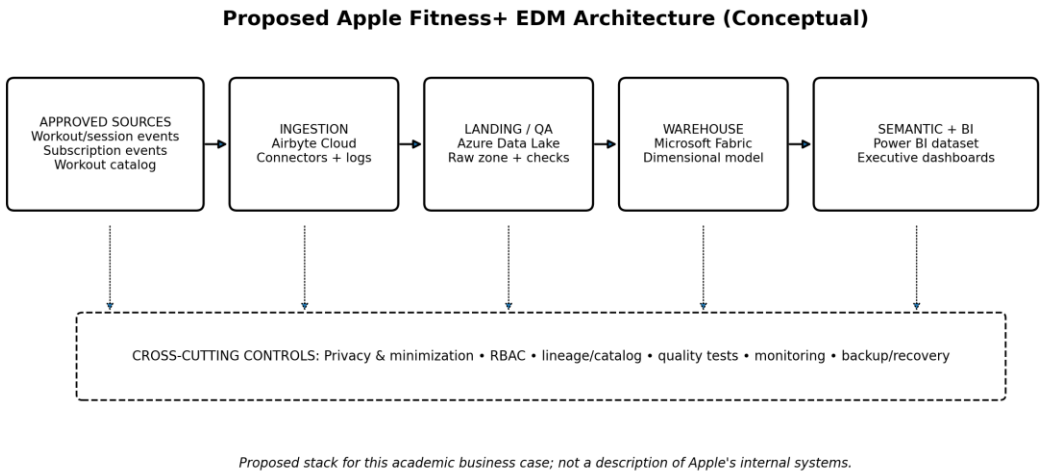


Figure 2

Proposed EDM Architecture

Note. Tool choices are a conceptual recommendation, not a description of Apple's internal technology.

Staffing and Implementation Plan

A lean pilot would use a sponsor/product owner (0.1 FTE), data engineer (0.5–1.0 FTE), BI developer/analyst (0.5 FTE), and part-time security/privacy and source-system owners. These are planning estimates, not Apple staffing facts. The ten-week timeline is: Weeks 1–2, discovery, source inventory, governance, and KPI definitions; Weeks 3–4, ingestion and raw-zone setup; Weeks 5–6, dimensional modeling and transformation; Weeks 7–8, Power BI measures, reconciliation, and user acceptance testing; Week 9, parallel run and training; Week 10, controlled launch and handoff. The proposed pipeline should run alongside existing reports until data totals and KPI definitions reconcile within agreed tolerances. Before cutover, management should approve a rollback plan, access roles, data-retention rules, pipeline runbooks, and incident escalation. Launch should proceed only after source owners sign off on data quality and business users validate the reports.

Initial Cost Analysis and Return on Investment

The following is an illustrative USD scenario for a ten-week pilot, not an Apple budget or vendor quote. Year 1 costs are estimated at $180,000: $120,000 for implementation labor, $20,000 for cloud, integration, and setup, and $40,000 for training, security review, and contingency. Annual operating costs are assumed to be $70,000 in Years 2 and 3. Estimated benefits are $120,000 in Year 1, $190,000 in Year 2, and $220,000 in Year 3, potentially reflecting analyst time recovered, less reconciliation, and better-informed retention experiments.

Three-year benefits total $530,000 and costs total $320,000, producing a $210,000 net benefit. ROI = (total benefits − total costs) ÷ total costs × 100 = 65.6%. Year 1 is negative by $60,000; cumulative break-even occurs during Year 2 if assumptions are achieved. Benefits are not guaranteed, and retention impact must be validated. Before approval, test a downside case with benefits 25% lower and costs 20% higher.

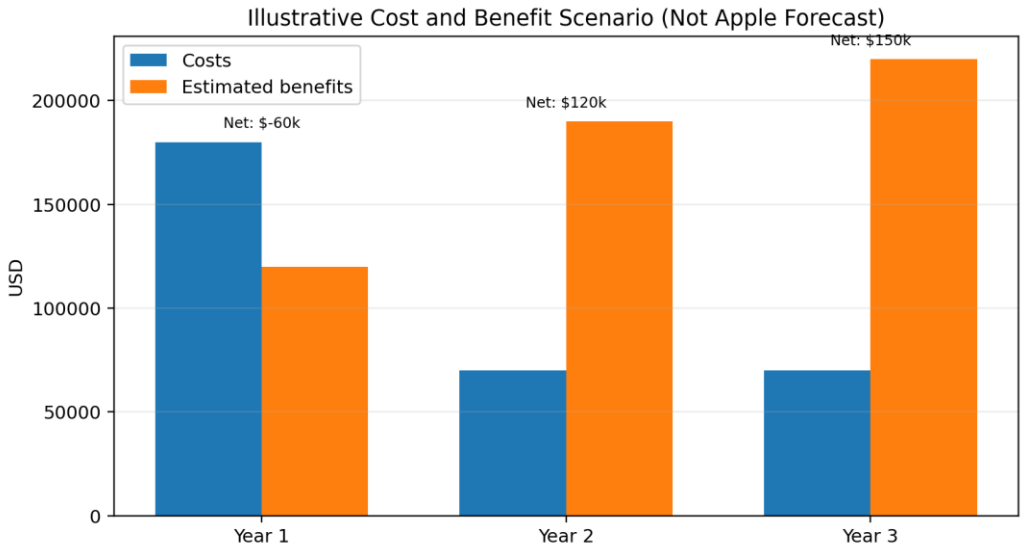


Figure 3

Illustrative Three-Year Cost and Benefit Scenario

Note. All amounts are modeled assumptions, not actual Apple financial data.

Benefits, Governance, and Maintenance

The proposed EDM capability would establish shared definitions for engagement, enable consistent cohort comparisons, reduce manual report reconciliation, and give leaders faster access to decision-ready information. Power BI could help identify segments for further investigation, but correlation between workouts and retention would not prove causation without an appropriate experiment or causal design.

After launch, the team should monitor pipeline failures and freshness, schema drift, quality exceptions, query performance and cost, access recertification, audit logs, backups and recovery, retention/deletion requests, and changes to KPI definitions. A named data owner should approve model changes, and the sponsor should review business value quarterly.

References

Airbyte. (n.d.). *Pricing.* https://airbyte.com/pricing

Apple. (2025a, January 3). *Apple Fitness+ unveils an exciting lineup of new ways to stay active and mindful in 2025.* Apple Newsroom. https://www.apple.com/newsroom/2025/01/apple-fitness-plus-unveils-an-exciting-lineup-of-new-ways-to-stay-active-in-2025/

Apple. (2025b, December 8). *Apple Fitness+ expands to 28 new markets.* Apple Newsroom. https://www.apple.com/newsroom/2025/12/apple-fitness-plus-expands-to-28-new-markets/

Kimball Group. (2003, January 1). *Fact tables and dimension tables.* https://www.kimballgroup.com/2003/01/fact-tables-and-dimension-tables/

Kimball Group. (2015, January 1). *Dimensional modeling techniques.* https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/kimball-techniques/dimensional-modeling-techniques/

Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling* (3rd ed.). Wiley.

Microsoft. (n.d.). *Power BI licensing for organizations.* Microsoft Learn. https://learn.microsoft.com/en-us/power-bi/enterprise/service-admin-purchasing-power-bi-pro